

Mechanical assessment

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Introduction

Changes in the mechanical properties of fascial tissues are of interest to researchers and clinicians. In order to assess such changes, invasive as well as non-invasive methods can be used. Invasive examination methods on a cellular and molecular level, in essence, include needle biopsy and immunohistochemistry. Non-invasive methods of assessing biomechanical tissue properties are employed in research as well. They have become increasingly relevant in clinical practice.

Rapid developments in technology are offering useful diagnostic and assessment tools that can be used to examine different physical and physiological features of fascial tissues. As considerable demand for diagnostic methods related to fascial tissue function has arisen, overviews of such methods have helped to gain better understanding and evaluation of the different methodologies (Wilke and Banzer, 2014; Zugel et al., 2018). This chapter draws from these current literature overviews and describes and evaluates various techniques, ranging from traditional and cost-neutral practices such as palpation to relatively new and costly methods such as ultrasound elastography (Table 21.1).

Assessment methods

Palpation

Examining tissue through touch traditionally plays an important role in clinical practice.

Assessments of properties such as tissue stiffness represent a frequent (although not always precise) tool in the detection of cancer, injuries and other diseases (Fowlkes et al., 1995). Palpation is cost-neutral and widely used by physicians and physiotherapists. However, many palpation methods, particularly when targeting soft tissue structures, appear to be subjective and lack reliability. For instance, investigations into the reproducibility of palpation in myofascial trigger point identification yielded conflicting results (Myburgh et al., 2008). To render the art of manual palpation more reliable, standardized assessment and training methodologies are necessary.

Palpation insight: assessing palpation skills

- Various methods and materials for assessing and teaching palpatory skills can be found in the literature. They range from simple self-built tools to more costly and technically engineered variants.
- Examples include the palpation of a coin under different layers of copy paper (Kamp et al., 2019), the use of grating domes (Mueller et al., 2014) as well as virtual reality simulations of mechanical properties of human tissue designed as an aid to teaching palpatory diagnosis (Howell et al., 2008).

Questionnaires/scales

In order to evaluate the effects of fascia-related interventions, questionnaires and scales related to the client's pain level are often used. Such

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Table 21.1 Overview of diagnostic methods to examine fascial tissues

Method	Assessment target	Advantages	Disadvantages
Palpation	Stiffness, elasticity, shearing mobility, as well as other tissue parameters	- Cost-effectiveness - Differentiation between structures partly possible - Psychosocial factors	- Subjective/depends on patient's and examiner's experience - Differentiation between structures limited - Palpation of deeper structures difficult
Questionnaires/scales	Perception of pain	- Cost-effectiveness - Generally reliable	- Differentiation between structures limited - Subjective results
Algometry	Perception of pain	- Semi-quantifiable results - Reliable	- Differentiation between structures only limited - Good applicability only with superficial structures
Indentometry	Stiffness and elasticity	- Quantification of results possible - Reliable	- Limited depth - Differentiation between structures only limited
Myometry	Stiffness	- Quantification of results possible - Reliable	- Limited depth - Differentiation between structures not possible
Electrical bioimpedance	Hydration changes	High sensitivity	- Reliability data are lacking - Validity for smaller body regions are lacking
Ultrasound	Thickness of layers, tendon elongation, relative shearing motion of adjacent layers	- Differentiation between structures possible - Reliable - Relatively cost-efficient	Lack of standardization regarding selection of exact viewing angles
Ultrasound-elastography	Stiffness	- Differentiation between structures possible - Quantification of results possible - Reliable	- Lack of standardization - Frequent appearance of artefacts - High acquisition cost

scales reflect the client's perception of the therapeutic effect of an intervention. While easy to use, they are an inherently subjective measurement tool and do not allow for conclusions related to a specific anatomical structure. Widely used scales include the Visual Analogue Scale (VAS), the Numerical Rating Scale (NRS) as well as the Verbal Rating Scale (VRS) (Figure 21.1; Williamson and Hoggart, 2005)

Algometry

The use of the Pressure Algometer (Park et al., 2011) allows the clinician/trainer not only to test tissue sensitivity, but also to evaluate the

client's pain threshold in a specific body location. With the tip of the algometer, the clinician applies increasing amounts of pressure to a particular area, for example, a trigger point (Myburgh et al., 2008), and asks the client to signal when the pain is uncomfortable. By doing this, the client gets an impression of the scale of pain, from 0 to 10, with 10 representing unbearable pain. The clinician then repeats the procedure at several points around the problem area, before and after treatment, and records the results. Tenderness pressure is measured in pounds (lbs) and tissue penetration is measured in millimeters (mm).

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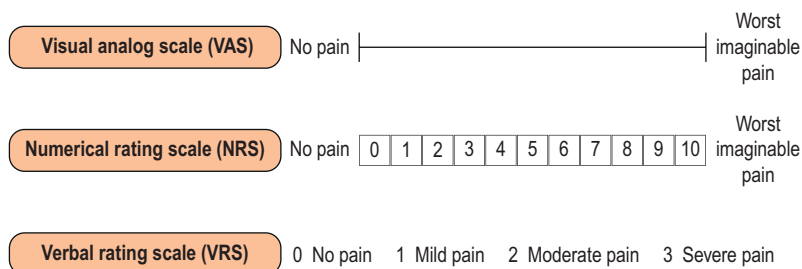


Figure 21.1

Common pain rating scales (Williamson and Hoggart, 2005).

Indentometry

Indentometry is an affordable and easy to handle technology to assess soft tissue stiffness. Hand-held devices such as the semi-electronic tissue compliance meter (STCM) or the IndentoPro consist of a modified algometer with a probe tip that is pressed into the tissue until a predefined penetration depth or force is reached. Stiffness is then specified as force per indentation depth (e.g. N/mm). While simple in handling, the STCM was reported to deliver reliable and valid measurements. As with other modes of assessment, reference values (e.g. for different body regions or musculoskeletal disorders) are still to be elucidated by future research (Wilke et al., 2018). The IndentoPro (Chemnitz, University of Technology; Figure 21.2) can provide measurements for both algometry and indentometry. It is easy to handle and has the advantage of providing a digital measurement.

Myometry

Hand-held myotonometer devices such as the Myotonometer or the MyotonPro aim to quantify the mechanical muscle properties. A testing probe is placed on the tissue so that short impulses generate oscillations in the tissue. The oscillation wave form reflects the visco-elastic properties of the tissue.

The MyotonPro is a myometer that gives numerical feedback on biomechanical tissue properties such as dynamic stiffness and



Figure 21.2

The IndentoPRO tool which can be used as an algometer (for the assessment of pressure pain sensitivity) and as indentometer (for stiffness assessment at different tissue depths).

Photo: fascialnet.com

elasticity. This method for measuring tissues close to the skin (up to depths of 1.5–2 cm) works by creating an external mechanical impulse. The

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muscle response is then recorded, in the form of an acceleration graph, and subsequently, the tone, elasticity and stiffness are computed. Reliability studies related to various body regions such as paraspinal muscles, gastrocnemius or Achilles tendon have reported that the MyotonPro can be used to reliably assess the mechanical properties of these tissues (Feng et al., 2018; Hu et al., 2018).

While the MyotonPro (Figure 21.3) has been used in various studies to assess the biomechanical properties of muscle tissue, the device generally does not only measure muscle tissue itself, but assesses various tissue layers. It is therefore important to note that the parameters quantified with this type of diagnostic, similar to those of the indentometer, relate to different tissue layers.



Figure 21.3
Myometry with MyotonPRO.

Research highlight: MyotonPro in space

In April 2018, a MyotonPro device was successfully launched to the International Space Station (ISS) for monitoring astronauts' muscle health and physical condition. The study funded by the European Space Agency, German Aerospace Center and UK Space Agency is called "Myotones—Muscle Tone in Space". It is the first time muscle tone assessment became technically possible in space. Staying fit through the space mission allows astronauts to return to normal life on earth as smoothly as possible. Measuring muscle tone with the MyotonPro is a crucial device to test muscle adaptation in space as part of the astronauts' daily routine. Scientists at the Charité University Berlin analyze the data on how muscles change during and after time on the ISS.

Electrical bioimpedance

Tissue hydration is frequently used as a hypothetical explanation for the benefits of many different types of bodywork. Schleip et al. (2012) showed that an increase in tissue stiffness may, at least, be partially caused by a temporarily altered matrix hydration.

Electrical bioimpedance is a fast, safe and inexpensive method using portable devices. The method has been used to assess nutritional status, cellular health and integrity as well as body composition (Martins et al., 2020). In sports, bioelectrical impedance analysis is used to determine the body composition and related hydration changes. It has been shown that athletic performance is influenced by and dependent on the distribution and total amount of fat-free mass and body fat. Bioimpedance has also been used to detect hydration changes before and after treatment and to help to determine influences that might change fluid distribution such as positioning. Lower impedance was shown along collagenous bands. Along these bands, some of the Traditional Chinese Medicine

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meridians could be represented. It is, therefore, of great interest to have a measurement tool that detects fluid distribution within the body.

The measured bioimpedance is the sum of the ohmic resistance of all fluids in the body, which is called resistance, and the capacitive resistance due to body cells, which is called reactance. Compared with muscle and blood, bone and fat are poorly conductive in the body. Therefore, if there is more bone and fat, the body is less conductive.

In electrical bioimpedance, several measurement methods can be performed using different frequencies and mathematical models. The use of different frequencies is necessary as they are required to estimate the various parameters such as fat-free mass or changes in the extra cellular water.

Whole-body impedance measurement is typically performed in supine position with four surface electrodes placed on one side of the body. Two current source electrodes are placed on the backside of the foot and hand and two detecting electrodes are positioned on the backs of the wrist and ankle (Foster and Lukaski, 1996; Kyle et al., 2004). A minimum distance of 3 cm should be kept between the current and the detecting electrode (Figure 21.4). While segmental measurements of body regions are possible and have advanced substantially in recent years, data on the reliability and validity of measurements in smaller body regions are still a matter of controversy. For instance, segmental impedance analysis tends to underestimate fat-free mass and overestimate fat mass (Jaffrin and Morel, 2008; Ward, 2012).

Ultrasound

Except for medical doctors, up until very recently, clinicians and coaches within the fields of sports and movement therapy rarely used any ultrasound equipment for their assessments.

However, technological advances have made portable ultrasound equipment an increasingly useful diagnostic tool within this field (Figure 21.5). Ultrasound technology has, for instance, been used to assess muscle activation, thickness of fascial tissues and “shear strain” between tissue layers.



Figure 21.4

Standard electrode position for whole-body impedance measurement.

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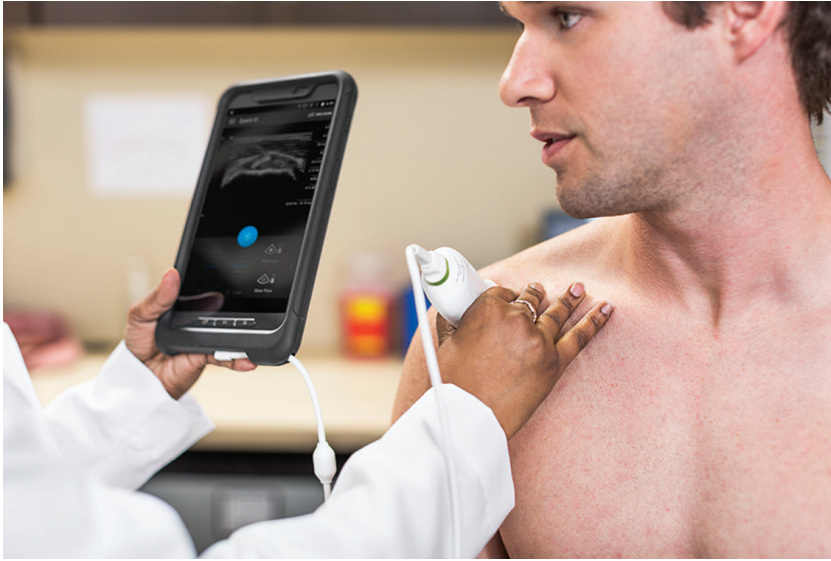


Figure 21.5

Use of ultrasound to visualize and assess fascial properties. Use of portable ultrasound allows for the measurement of thickness of various fascial membranes positioned close to the surface.

Photo: Philips Health System

Practitioners focusing on segmental stabilization, via muscular activation training, have successfully used ultrasound in their daily practices to assess proper (or lacking) activation of muscle layers related to “core stability”, such as the transversus abdominis or the multifidus muscle (Hodges et al., 2003).

Ultrasound may also be used effectively to measure the thickness of fascial tissues (Figure 21.5).

Research highlights: tissue thickness from head to heel

- An impressive clinical study by Stecco et al. (2014) showed that in patients with chronic neck pain, the thickness of the sternocleidomastoid fascia corresponded to the amount of pain and disability, and also to the degree of therapeutic improvement after a myofascial manipulation treatment. Furthermore, a fascial thickness of 1.5 mm was found to be a reliable cut-off point in this study for the diagnosis of fascia-generated neck pain.

- Langevin et al. (2009) reported assessed connective tissue structures in the lumbar region with ultrasound. Connective tissue structure was altered in a group of subjects with chronic or recurrent low back pain. The perimuscular tissue layer was ~25% thicker in the low back pain group compared with controls without low back pain. Differences in age, sex, body mass index or activity level did not influence the results. The differences found may have their cause in genetic factors, abnormal movement patterns or chronic inflammation.
- Holowka et al. (2019) used portable ultrasound technology in Kenya and the USA to assess foot callus thickness. They found that people who frequently walk barefoot have thicker calluses than those who usually use footwear. However, callus thickness did not trade-off protection for the ability to perceive tactile stimuli. Furthermore, callus thickness did not affect how hard the feet strike the ground during walking.

A prominent study by Langevin et al. (2011) showed, via ultrasound assessment, that patients

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with chronic low back pain tend to express significantly less shear strain (corresponding to the ability to slide in relation to one another) between different layers of their lumbar fascia compared with a pain-free control group. This type of assessment has also been used to research the effect of foam rolling on the low back region in healthy young adults (Frieder et al., 2019). Foam rolling significantly increased the mobility of the thoracolumbar fascia in the studied population.

Some studies use ultrasound to visualize myofascial trigger points. While the value of an ultrasound diagnostic for trigger point detection is still a matter of debate (Wilke and Banzer, 2014), most of the available studies found it to be a suitable method of diagnosis, and novel methods to measure myofascial trigger points using ultrasound imaging are still being developed (Jafari et al., 2018).

Ultrasound-elastography

A fairly new development with increasing application in musculoskeletal research is ultrasound elastography. This technology was created in the 1990s but has only recently been applied to soft tissue imaging of muscles. This cutting-edge technology provides images reflecting the hardness of the analyzed region (Zugel et al., 2018).

A mechanical vibration of the respective tissue is induced, for which the detected resonance frequency allows an assessment of the tissue stiffness. The softer a fascial tissue is, the slower its induced vibratory frequency. This non-invasive technology is already used to assist in the detection of liver fibrosis, breast cancer and prostate cancer as well as plantar fibrosis (Sconfienza et al., 2013). Furthermore, its real-time measurement of muscle stiffness can aid the diagnosis and rehabilitation of musculoskeletal injuries and chronic myofascial pain and also help to monitor intervention outcomes (Brandenburg et al., 2014).

Ultrasound elastography insight: strain versus shear wave elastography

- **Strain elastography:** the examiner compresses the ultrasound transducer against the body surface (e.g. muscle) manually. Softer tissue shows more deformation (and therefore experiences larger strain) than stiffer tissue. This is shown in an elastogram and represents a qualitative assessment. Translating this into a quantitative measure is challenging. One semi-quantitative method is the strain ratio. The elastography software first calculates the average strain in the normal area and then divides it by the average strain in the diseased area (e.g. a myofascial trigger point). However, these techniques are not in reality quantitative, but rather a relative assessment of tissue stiffness.
- **Shear wave elastography:** this technique uses shear waves to assess tissue stiffness. The shear waves travel perpendicularly through tissue to the direction of the particle motion. As the stiffness of underlying tissue increases, the shear wave speed increases. When looking at shear wave elastography studies, it is important to note that both the Young and shear modulus have been used in reporting outcomes, with the latter being the more accurate measure. The two measures differ slightly in their theoretical assumptions, but display good agreement with one other. The Young modulus can be converted to the shear modulus by dividing by 3 (Brandenburg et al., 2014).

Unfortunately, the prices of new equipment are still comparatively high. However, the respective technological industry has started to target physiotherapists, and others, as a new customer group. This has created the necessity to develop more affordable versions, with impressive imaging quality incorporated into smaller therapeutic practices.

Apart from the acquisition cost, the technology has a couple of other limitations. The amount of

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pressure to be applied to the tissue, the distance between the probe and the tissue, as well as the size of the region of interest, need to be carefully selected and standardized, as these factors can influence the results and their interpretation. Artefacts such as fluctuant changes at the edges of the elastogram and at the borders of thin structures can occur. Familiarity with such artefacts is important for the interpretation of results. Furthermore, there are currently many different techniques and processing algorithms in use. Accordingly, the findings and artefacts may be highly dependent on the technique used (Drakonaki et al., 2012).

Application and benefits

Measuring biomechanical tissue qualities before and after treatment or training has many benefits. The most obvious advantage is being able to track client progress. The practitioner or trainer has a better tool for communication with

the client by being able to demonstrate objective evidence of treatment progress. This means less reliance is placed on the patient's subjective accounts of improvement (or lack thereof), and more on objective assessment tools, which measure an array of parameters.

Objective assessment allows for monitoring the quality and effectiveness of treatments. Clinicians can improve techniques through self-assessment, which leads to better methodologies and advancements in this field of health.

As in any field characterized by rapid technological advancements, evidence-based recommendations for clinical application will develop and then be refined within the next few years. Meanwhile, the enthusiastic clinician should combine methods and interpret findings with care, bearing in mind that factors such as the operator's experience or measuring technique may influence the results.

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Further reading

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